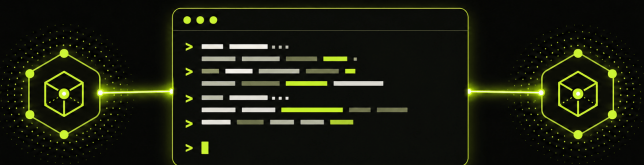


p2p-netcat

p2p-netcat web

A terminal between two peers.
Right in your browser.



Netcat addressed by PeerId

One cryptographic address for terminal, TCP, UDP, and proxy access across NAT.

IDENTITY

PeerId instead of IP

ROUTE

First authenticated route

SESSION

TCP · UDP · PTY · SOCKS

Your home server has no public address – but you still need SSH.

WITHOUT P2P-NETCAT

The network edge is not yours

Laptop

Home router

CGNAT

Port forwarding is unavailable. Dynamic DNS does not help. A hosted VPN adds another control plane.

WITH P2P-NETCAT

The server keeps one identity

12D3KooW...YXdUfbR9

discover · authenticate · connect

Run a listener at home. From anywhere, dial its PeerId and forward local port 2222 to SSH.

A full Internet tunnel should still work when the gateway is behind NAT.

TRAVEL · LAPTOP

WireGuard

Endpoint

127.0.0.1:15182

p2p-nc

keeps carrier

outside wg0

policy route protects the carrier

UNTRUSTED NETWORK

hotel · airport · mobile hotspot

no inbound port required

HOME · GATEWAY

p2p-nc

UDP listener

→ UDP :51820

WireGuard

home gateway

10.66.66.1/24

FORWARD + MASQUERADE

ONE ROUTE

bring up carrier

wg-quick up wg0

public IP = home

The P2P carrier reaches the home peer first; WireGuard then routes all Internet traffic through it.

The same need keeps repeating: reach a private service without owning its network edge.

Home lab

SSH · NAS

TCP

Family support

RDP · web UI

TCP

Remote work

PostgreSQL · dev API

TCP

Private browsing

SOCKS5h

SOCKS

Legacy VPN

OpenVPN

STREAM

Full tunnel

WireGuard

UDP

One stable PeerId replaces per-network IP addresses, port mappings, and temporary VPN control planes.

PeerId solves identity. The route still has to be found.

CLASSIC FORWARDING

The IP changes – access breaks

192.0.2.10:22

NAT / CGNAT

DNS, port forwarding, a VPN concentrator, or manually operated infrastructure.

P2P-NETCAT

Identity stays stable

12D3KooW...YXdUfbR9

mDNS · DHT · signaling · relay hints

PeerId selects the right peer; discovery and route racing select a reachable path.

The first route to complete authentication wins.

CLIENT

CLI or browser PWA
PeerId + logical port

PUBLIC ROUTING

mDNS · GossipSub · DHT
NATIVE SIGNALING
Nostr + WebTorrent

LIBP2P PLANE

QUIC · WebRTC Direct · WSS · TCP · Relay
PION WEBRTC
ICE/STUN · ordered DataChannel

AUTH GATE

Expected PeerId
admission handshake

LISTENER

Persistent Ed25519 identity

logical service
1...65535

SESSION AFTER AUTH

raw / exec
TCP forward
UDP frames
SOCKS
PTY / ConPTY

Relay-only keeps the PeerId local – and makes ‘no direct path’ enforceable.

DEFAULT ROUTING

Fastest authenticated route wins

direct TCP · QUIC

WebRTC · DHT

Direct and relayed candidates can race. Reachability is maximized, but the host may expose a direct route.

STRICT RELAY-ONLY

Only explicit Circuit Relay routes exist

LOCAL PEERID · E2E NOISE/TLS

TCP/WSS relay → p2p-circuit

Requires an explicit relay; disables DHT, mDNS, PubSub, direct QUIC/WebRTC and fails closed if the relay is unavailable.

A trusted WSS gateway moves the P2P host – and the trust boundary – to the VPS.

LOCAL DEVICE

p2p-nc-lite
PTY · TCP · UDP
local session owner
WSS agent
one outbound link
TCP 443
pairing + admission stay in the agent

TRUSTED GATEWAY

per-device hosted PeerId
full P2P route selection

REMOTE PEER

p2p-nc
ordinary client/listener
exact PeerId + port
Circuit Relay
shared WSS :443
no random ports
GATEWAY MAY SEE APPLICATION PLAINTEXT

ONE FIXED PORT

agent opens WSS
gateway hosts PeerId
P2P stream reaches peer
Data, EOF, reset, and flow credit are multiplexed; local -i still starts the PTY on the device.

One CLI covers terminals, streams, datagrams, and proxies.

RAW / EXEC

stdin/stdout
stream or command
byte stream
-e

TCP FWD

127.0.0.1:PORT
dial TCP target
connection
-p

UDP FWD

127.0.0.1:PORT/udp
connected UDP socket
packet boundaries
-u -p

SOCKS

SOCKS4/4a/5
CONNECT remotely
remote DNS
-S

PTY

interactive TTY
Unix PTY / ConPTY
resize + EOF
-i

LISTENER

logical port
multiple sessions
identity + policy
-l -k

Wire protocol compatibility

/p2p-netcat/1.0.0/<port> · /p2p-netcat/udp/1.0.0/<port>

Application bytes flow only after the expected PeerId is proven.

Offer

Client creates SDP

DataChannel

p2p-netcat-v2

Challenge

32 random bytes

Signature

Ed25519 + service

PeerId check

Exact match

AUTH_READY

Session allowed

AFTER AUTH

256

KiB

end-to-end flow window

p2p-netcat · PeerId access across NAT

120 sec

reconnect on the same logical stream

UDP packet boundaries survive even over TCP/WSS/relay.

LOCAL SOURCE

127.0.0.1:15182
association per endpoint

00 84

uint16 BE length
exact UDP payload bytes

P2P STREAM

authenticated + encrypted
ordered reliable carrier

FIXED UDP TARGET

WireGuard · DNS · game service
exact reply to original source

Native WebRTC

ICE/STUN direct

QUIC / WebRTC Direct

direct UDP path

TCP / WSS

UDP-over-stream

Circuit Relay v2

fallback path

Tor + TCP relay

restricted networks

b2b-netcat · PeerId access across NAT

Fallback cost: ordered delivery can introduce head-of-line blocking.

The carrier stays outside the default route, so the tunnel cannot consume itself.

NAT A • CLIENT

WireGuard

Endpoint

127.0.0.1:15182

p2p-nc

UDP carrier

special UID

ip rule uidrange → lookup main

PUBLIC INTERNET

Nostr/WebTorrent signaling + STUN/ICE

application packets travel over selected P2P carrier

NAT B • GATEWAY

p2p-nc

UDP listener

→ UDP :51820

WireGuard

wg0 gateway

10.66.66.1/24

FORWARD + MASQUERADE

E2E CHECK

two NAT namespaces

WireGuard handshake

HTTP egress = full-tunnel-ok

Start the carrier first, then wg-quick up wg0 – policy routing protects DNS, signaling, STUN, and reconnect.

One primitive operation powers many working tools.

OpenSSH

ProxyCommand / :2222

TCP

PostgreSQL

:15432 → :5432

TCP

RDP

:13389 → :3389

TCP

SOCKS5h

remote DNS

SOCKS

OpenVPN

tcp-client

STREAM

WireGuard

full Internet tunnel

UDP

The logical port selects the P2P service; it does not have to match the local TCP/UDP port.

One installation command; four delivery surfaces.

VERIFIED DEPLOY

curl -fsSL .../deploy.sh | bash
detects OS + architecture
downloads the release
verifies SHA256SUMS
installs p2p-nc + pnc

RELEASE BINARIES

Linux · macOS · Windows · Android
amd64 / arm64

GHCR CONTAINER

ghcr.io/santaklouse/go-p2p-netcat
non-root · /config identity

GO INSTALL

cmd/p2p-nc + cmd/pnc
CGO is not required

BROWSER PWA

Static

GitHub Pages

React UI

Web Worker

IndexedDB routes

p2p-netcat · PeerId access across NAT

Native WebRTC

Tests target the exact failure boundaries introduced by the architecture.

NATIVE WEBRTC SOAK

60 / 60

workload iterations

1 152 MiB

transferredBytes in a fresh JSON report

Pion DataChannel · SHA-256 · reconnect

CI COVERAGE

Go tests

Linux · macOS · Windows

Race detector

Linux · ./...

Full tunnel

2 NAT + WireGuard + HTTP

Docker

buildx + runtime checks

WebRTC soak

weekly Ubuntu + macOS

WHAT SOAK DOES NOT PROVE

public Nostr/tracker availability

intercontinental latency

every NAT and firewall type

background browser tabs and sleep

100% relay-free connectivity

p2p-netcat · PeerId access across NAT

Three command pairs cover the highest-value use cases.

SSH THROUGH A LOCAL PORT

```
server$ p2p-nc -l -k -d 127.0.0.1 -p 22 22022
client$ p2p-nc -p 2222
12D3KooWQ3uxpHgjdKE6vGmvzKS8RPbxUDLwJ7XCLaD6YXdUfbR9
22022
client$ ssh -p 2222 alice@127.0.0.1
```

TCP PORT FORWARDING

```
server$ p2p-nc -l -k -d 127.0.0.1 -p 5432 25432
client$ p2p-nc -p 15432
12D3KooWQ3uxpHgjdKE6vGmvzKS8RPbxUDLwJ7XCLaD6YXdUfbR9
25432
```

WIREGUARD CARRIER FOR A FULL TUNNEL

```
server$ p2p-nc -u -l -k -d 127.0.0.1 -p 51820 35182
client$ sudo wireguard-full-tunnel.sh --
/usr/local/bin/p2p-nc -u --udp-idle-timeout 0 \
    -p 15182
12D3KooWQ3uxpHgjdKE6vGmvzKS8RPbxUDLwJ7XCLaD6YXdUfbR9
35182
client$ sudo wg-quick up wg0
```


Use p2p-netcat when identity should outlive the address.

PeerId is the stable entry point. Route racing and fallback are the delivery mechanism.

github.com/santaklouse/go-p2p-netcat

GOOD FIT

access to home and edge hosts

SSH and private TCP services

UDP/WireGuard across NAT

temporary tunnels without a control plane

PLAN B REQUIRED

symmetric NAT or blocked UDP

guaranteed enterprise ingress

unsupported SOCKS UDP ASSOCIATE

TURN · Circuit Relay v2 · reachable TCP/WSS route